Database Design

7.1 Overview

The AI-Powered Taxi Dispatch & Management Platform uses a **MongoDB NoSQL document database** designed with a **multi-tenant architecture** to ensure secure data isolation between taxi companies. The database supports real-time booking, AI processing, dispatch operations, GPS tracking, payment management, reporting, and audit logging while maintaining high performance and scalability.

The database is schema-modeled (using embedding and referencing strategies) to reduce redundancy, maintain data integrity, and support efficient querying for operational and analytical workloads.

7.2 Database Objectives

The database is designed to:

- Store all operational and business data securely.
- Support multiple taxi companies (Multi-Tenant).
- Maintain data integrity using document references and schema validation.
- Handle thousands of concurrent bookings.
- Support real-time GPS tracking.
- Store AI conversations and learning data.
- Manage payment transactions and invoices.
- Maintain audit logs and call recordings.
- Support reporting and analytics.
- Ensure backup and disaster recovery.

7.3 Database Architecture

Database Type: MongoDB

Architecture: Multi-Tenant Document-Oriented Database

Caching: Redis

Storage: Cloud Object Storage (Call recordings & documents)

7.4 Core Database Collections

Company Management

Companies

Stores taxi company information.

Field	Type
CompanyID (PK)	UUID
CompanyName	VARCHAR
Email	VARCHAR
Phone	VARCHAR
Address	TEXT
TimeZone	VARCHAR
Status	BOOLEAN
CreatedAt	TIMESTAMP

User Management

Users

Stores all platform users.

Field	Type
UserID (PK)	UUID
CompanyID (FK)	UUID
Role	ENUM
FullName	VARCHAR
Email	VARCHAR
Phone	VARCHAR
PasswordHash	TEXT
Status	BOOLEAN
LastLogin	TIMESTAMP

Customer Management

Customers

Field	Type
CustomerID (PK)	UUID
CompanyID (FK)	UUID
FirstName	VARCHAR
LastName	VARCHAR

Mobile	VARCHAR
Email	VARCHAR
DefaultAddress	TEXT
PreferredPayment	VARCHAR
CreatedAt	TIMESTAMP

Driver Management

Drivers

Field	Type
DriverID (PK)	UUID
CompanyID (FK)	UUID
UserID (FK)	UUID
LicenseNumber	VARCHAR
VehicleID (FK)	UUID
Status	ENUM
OnlineStatus	BOOLEAN
Rating	DECIMAL

Vehicle Management

Vehicles

Field	Type
VehicleID (PK)	UUID
CompanyID (FK)	UUID
RegistrationNumber	VARCHAR
VehicleType	VARCHAR
Capacity	INTEGER
Model	VARCHAR
Year	INTEGER
Status	BOOLEAN

Booking Management

Bookings

Field	Type
BookingID (PK)	UUID
CompanyID (FK)	UUID
CustomerID (FK)	UUID
PickupAddress	TEXT
DestinationAddress	TEXT

BookingType	ENUM
BookingTime	TIMESTAMP
Status	ENUM
EstimatedFare	DECIMAL

Trip Management

Trips

Field	Type
TripID (PK)	UUID
BookingID (FK)	UUID
DriverID (FK)	UUID
VehicleID (FK)	UUID
StartTime	TIMESTAMP
EndTime	TIMESTAMP
Distance	DECIMAL
TotalFare	DECIMAL
TripStatus	ENUM

Payment Management

Payments

Field	Type
PaymentID (PK)	UUID
TripID (FK)	UUID
CustomerID (FK)	UUID
PaymentMethod	VARCHAR
Amount	DECIMAL
TransactionID	VARCHAR
PaymentStatus	ENUM
PaidAt	TIMESTAMP

Driver Payouts

DriverPayouts

Field	Type
PayoutID (PK)	UUID
DriverID (FK)	UUID
TripID (FK)	UUID
Amount	DECIMAL
Status	ENUM

PaidDate	TIMESTAMP

Pricing Engine

PricingRules

Field	Type
RuleID (PK)	UUID
CompanyID (FK)	UUID
BaseFare	DECIMAL
PerKmRate	DECIMAL
WaitingCharge	DECIMAL
NightSurcharge	DECIMAL
AirportFee	DECIMAL

Service Zones

Zones

Field	Type
ZoneID (PK)	UUID
CompanyID (FK)	UUID
ZoneName	VARCHAR

ZoneType	VARCHAR
PolygonCoordinates	JSON

GPS Tracking

DriverLocations

Field	Type
LocationID (PK)	UUID
DriverID (FK)	UUID
Latitude	DECIMAL
Longitude	DECIMAL
Speed	DECIMAL
RecordedAt	TIMESTAMP

Notifications

Notifications

Field	Type
NotificationID (PK)	UUID
UserID (FK)	UUID
NotificationType	VARCHAR

Message	TEXT
Status	ENUM
SentAt	TIMESTAMP

AI Management

AIBookings

Stores AI-generated booking requests.

AITrainingData

Stores dispatcher corrections and AI learning feedback.

AIConversationLogs

Stores AI conversations for analysis and model improvement.

Call Recording

CallRecordings

Field	Type
RecordingID (PK)	UUID
CompanyID (FK)	UUID
CallerNumber	VARCHAR
CallType	ENUM
RecordingURL	TEXT

Duration	INTEGER
CreatedAt	TIMESTAMP

Audit Logging

AuditLogs

Field	Type
AuditID (PK)	UUID
UserID (FK)	UUID
Action	VARCHAR
Module	VARCHAR
IPAddress	VARCHAR
Timestamp	TIMESTAMP

7.5 Entity Relationships

- One **Company** has many **Users**, **Drivers**, **Customers**, **Vehicles**, **Bookings**, and **Pricing Rules**.
- One **Customer** can create many **Bookings**.
- One **Booking** generates one **Trip**.
- One **Driver** can complete many **Trips**.
- One **Vehicle** can be assigned to many **Trips** over time.
- One **Trip** can have one or more **Payments** and a related **Driver Payout**.
- **Audit Logs**, **AI Conversations**, **Notifications**, and **Call Recordings** are linked to their respective users or companies for traceability.

7.6 Database Security

- Multi-Tenant Data Isolation
- Reference-Based Data Integrity (Application/Schema-Enforced)
- Indexed Fields (_id and Query-Optimized Indexes)
- Encrypted Sensitive Data
- Role-Based Database Access
- Automated Backups
- Audit Logging
- Soft Delete Support
- Point-in-Time Recovery (PITR)

7.7 Database Design Summary

The proposed database consists of approximately **18–22 core collections**, supporting all major modules including company management, user administration, customer and driver management, bookings, trips, dispatch, GPS tracking, AI processing, IVR interactions, payments, reporting, notifications, audit logs, and call recordings. The schema is optimized for scalability, high availability, and enterprise-grade performance while supporting future expansion without significant structural changes.